Monotone Comparative Statics
Finite Data and GARP

Econ 2100 Fall 2018

Lecture 7, September 19

- Problem Set 3 was due by 10:30am today

Outline

1. Comparative Statics Without Calculus
2. Supermodularity and Single Crossing
3. Topkis and Milgrom&Shannon's Theorems
4. Finite Data

Comparative Statics Without Calculus

Remark

- Let $x^*(q) = \arg\max f(x, q),$ subject to $q \in \Theta$, $x \in S(q)$

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2. Conclusions hold only in a neighborhood of the optimum.
3. Results are dependent on the functional form used for the objective function.
   1. In particular, IFT gives cardinal results that depend on the assumptions on f.

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Objectives

- With monotone comparative statics, we have results about "changes" that:
  - do not need calculus;
  - are not necessarily local;
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- If the optimum is not unique, then $x^*(q)$ is a correspondence, but what does it mean for a correspondence to be increasing?

Strong Set Order

- Ranking real numbers is easy, but how can we express the idea that one set is bigger than another set?

Definition

For two sets of real numbers A and B, define the binary relation $\geq_s$ as follows:

$$A \geq_s B \quad \text{if} \quad \text{for any } a \in A \text{ and } b \in B, \ \min\{a, b\} \in B \quad \text{and} \quad \max\{a, b\} \in A$$

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Example

Suppose $A = \{1, 3\}$ and $B = \{0, 2\}$. Then, A is not greater than or equal to B in the strong set order.

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Prove that if $g(\cdot)$ is non-decreasing and $\min g(x)$ exists for all x, then $\min g(x)$ is non-decreasing.

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- Assume existence of a solution, but not uniqueness.

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The function $f : \mathbf{R}^2 \to \mathbf{R}$ is supermodular in $(x; q)$ if

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Question 1, Problem Set 4.

Show that supermodularity is equivalent to the property that

$$q' > q \qquad \text{implies} \qquad f(x; q') - f(x; q) \text{ is non-decreasing in } x.$$

Differentiable Version of Supermodularity

- When f is smooth, supermodularity has a characterization in terms of derivatives.

Lemma

A twice continuously differentiable function $f : \mathbf{R}^2 \to \mathbf{R}$ is supermodular in $(x; q)$ if and only if $D_{xq} f(x; q) \geq 0$ for all $(x; q)$.

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  - $q' > q$ implies $f(x; q') - f(x; q)$ is non-decreasing in x

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Theorem (Easy Topkis' Monotonicity Theorem)

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  - By supermodularity, $f(\max\{x, x'\}; q') - f(x'; q') \geq 0$,
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- Now show that $\min\{x, x'\} \in x^*(q)$

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If f is supermodular in $(x; q)$, then $x^*(q) = \arg\max f(x, q)$ is non-decreasing.

- Supermodularity is sufficient to draw comparative statics conclusions in optimization problems without other assumptons.
- Topkis' Theorem follows from the IFT whenever the standard full-rank condition in the IFT holds.
  - At a maximum, if $D_{xx} f(x^*, q) \neq 0$, it must be negative (by the second-order condition), hence the IFT tells you that $x^*(q)$ is strictly increasing.

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Profit Maximization Without Calculus

- A monopolist chooses output q to solve $\max p(q)q - c(q, \mu)$.
  - $p(\cdot)$ is the demand (price) function
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- Notice that supermodularity says that for all $q' > q$, $-c(q'; \mu) + c(q; \mu)$ is nondecreasing in μ.
  - in other words, the marginal cost is decreasing in μ.
- Conclusion: by Topkis' theorem q^* is nondecreasing as long as the marginal cost of production decreases in the technological progress parameter μ.

Single-Crossing

- In constrained maximization problems, $x \in S(q)$, supermodularity is not enough for Topkis' theorem.

Definition

The function $f : \mathbf{R}^2 \to \mathbf{R}$ satisfies the single-crossing condition in $(x; q)$ if for all $x' > x$, $q' > q$

$$f(x'; q) - f(x; q) \geq 0 \quad \text{implies} \quad f(x'; q') - f(x; q') \geq 0$$

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Theorem

If f satisfies single crossing in $(x; q)$, then $x^*(q) = \arg\max_{x \in S(q)} f(x; q)$ is nondecreasing. Moreover, if $x^*(q)$ is nondecreasing in q for all constraint choice sets S, then f satisfies single-crossing in $(x; q)$.

Monotone Comparative Statics

n-dimensional choice variable and m-dimensional parameter vector

- Next, we generalize to higher dimensions.

Definitions

Suppose $x, y \in \mathbf{R}^n$.

- The join of x and y is defined by
$$x \vee y = (\max\{x_1, y_1\}, \max\{x_2, y_2\}, \ldots, \max\{x_n, y_n\}).$$
- The meet of x and y is defined by
$$x \wedge y = (\min\{x_1, y_1\}, \min\{x_2, y_2\}, \ldots, \min\{x_n, y_n\}).$$

- Draw a picture.

Strong Set Order

- We generalize the strong set order definition to $\mathbf{R}^n$.

Definition (Strong set order in $\mathbf{R}^n$)

The binary relation $\geq_s$ is defined as follows: for $A, B \subset \mathbf{R}^n$,

$$A \geq_s B \quad \text{if} \quad \text{for any } a \in A \text{ and } b \in B, \; a \wedge b \in B \quad \text{and} \quad a \vee b \in A$$

- The meet is in the smaller set, while the join is in the larger set.
- One uses this to talk about non-decreasing $\mathbf{R}^n$-valued correspondences.

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- We generalize the strong set order definition to $\mathbf{R}^n$.

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- We look at functions $f(\mathbf{x}; \mathbf{q})$ where the first argument represents the endogenous variables and the second represents the parameters.

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2. Suppose $f(\mathbf{x};\mathbf{q})$ is twice differentiable in x and $\frac{\partial^2 f}{\partial x_i x_j} > 0$ for all $i, j = 1, \ldots, n$ with $i \neq j$. Then f is quasi-supermodular in $\mathbf{x}$.

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2. $$f(\mathbf{x}; \mathbf{r}) > f(\mathbf{y}; \mathbf{r}) \quad \Rightarrow \quad f(\mathbf{x}; \mathbf{q}) > f(\mathbf{y}; \mathbf{q}).$$

- The "marginal return" $f(\mathbf{x}; \cdot) - f(\mathbf{y}; \cdot)$ as a function of the second argument can cross 0 at most once.
- The single-crossing property is an ordinal property; for differentiable functions there is a sufficient condition for single-crossing.

Exercise

1. Prove that if f satisfies the single-crossing property, then $h \circ f$ satisfies the single-crossing property for any strictly increasing $h : f(\mathbf{R}^n \times \mathbf{R}^m) \to \mathbf{R}$.

Single-Crossing Property

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1. Prove that if f satisfies the single-crossing property, then $h \circ f$ satisfies the single-crossing property for any strictly increasing $h : f(\mathbf{R}^n \times \mathbf{R}^m) \to \mathbf{R}$.
2. Suppose $f(\mathbf{x}; \mathbf{q})$ is twice differentiable and $\frac{\partial^2 f}{\partial x_i x_j} > 0$ for all $i = 1, \ldots, n$ and $j = 1, \ldots, m$. Then f satisfies the single-crossing property.

Monotone Comparative Statics

Theorem (easy Milgrom and Shannon)

Let $f : \mathbf{R}^n \times \mathbf{R}^m \to \mathbf{R}$. *Define* $x^*(\mathbf{q}) = \arg\max_{x \in R^n} f(\mathbf{x}; \mathbf{q})$. *Suppose* $|x^*(\mathbf{q})| = 1$ *for all* $\mathbf{q}$ *and* $f(\mathbf{x}; \mathbf{q})$ *is quasi-supermodular in its first argument and satisfies the single-crossing property. Then*

$$\mathbf{q} \geq \mathbf{r} \Rightarrow x^*(\mathbf{q}) \geq x^*(\mathbf{r}).$$

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Proof.

Suppose $\mathbf{q} \geq \mathbf{r}$. Then:

$$f(x^*(\mathbf{r}); \mathbf{r}) \geq f(x^*(\mathbf{q}) \wedge x^*(\mathbf{r}); \mathbf{r}) \quad \text{by definition of } x^*(\mathbf{q})$$

$$\Rightarrow \quad f(x^*(\mathbf{q}) \vee x^*(\mathbf{r}); \mathbf{r}) \geq f(x^*(\mathbf{q}); \mathbf{r}) \quad \text{by quasi-supermodularity in } \mathbf{x}$$

$$\Rightarrow \quad f(x^*(\mathbf{q}) \vee x^*(\mathbf{r}); \mathbf{q}) \geq f(x^*(\mathbf{q}); \mathbf{q}) \quad \text{by Single Crossing}$$

$$\Rightarrow \quad x^*(\mathbf{q}) \vee x^*(\mathbf{r}) = x^*(\mathbf{q}) \quad \text{since } |x^*(\mathbf{q})| \text{ equals 1}$$

$$\Rightarrow \quad x^*(\mathbf{q}) \geq x^*(\mathbf{r}) \quad \text{by Question 2, PS4}$$

□

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- This result can be extended to constrained optimization problems (Question 3, Problem Set 4), and to multi-valued optimizers (see Milgrom & Shannon).

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Main Idea

- We observe data and want to know if it could have been the result of maximizing a preference relation or a utility function.

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- We observe N consumption choices made by an individual, given her income and prices (also observable):

  $$\mathbf{x}^1, \mathbf{p}^1, w^1 \qquad \mathbf{x}^2, \mathbf{p}^2, w^2 \qquad \mathbf{x}^3, \mathbf{p}^3, w^3 \qquad ... \qquad \mathbf{x}^N, \mathbf{p}^N, w^N$$

  These satisfy:

  - $(\mathbf{x}^j, \mathbf{p}^j, w^j) \in \mathbf{R}^n_+ \times \mathbf{R}^n_{++} \times \mathbf{R}_+$ for all $j = 1, .., N$; and
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- In other words, we have finitely many observations on behavior.

- What conditions must these observations satisfy for us to conclude they are the result of the maximizing of a preference relation or a utility function?
- Answer: Something similar to, but stronger than, revealed preference.

An Example (from Kreps)

- Suppose we observe the following choices among 3 goods

	w	p	x
1	300	$(10, 10, 10)$	$(10, 10, 10)$
2	130	$(10, 1, 2)$	$(9, 25, 7.5)$
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Consequences of Local Non Satiation

- The following is slightly different from Full Expenditure.

Lemma

Suppose $\succsim$ *is locally non-satiated, and let* $\mathbf{x}^*$ *be an element of Walrasian demand (therefore* $\mathbf{x}^* \succsim \mathbf{x}$ *for all* $\mathbf{x} \in \{\mathbf{x} \in X : p \cdot \mathbf{x} \leq w\}$*). Then*

$$\mathbf{x}^* \succsim \mathbf{x} \qquad \text{when} \qquad p \cdot \mathbf{x} = w$$

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- The maximal bundle is strictly preferred to any bundle that costs less.

Proof.

The first part is immediate.
For the second part note that if $p \cdot \mathbf{x} < w$ by local non satiation (and continuity of $p \cdot \mathbf{x}$) there exists some $\mathbf{x}'$ such that $\mathbf{x}' \succ \mathbf{x}$ and $p \cdot \mathbf{x}' \leq w$.
Thus $\mathbf{x}'$ is affordable and $\mathbf{x}^* \succsim \mathbf{x}' \succ \mathbf{x}$ as desired. □

Example Continued

Consumer's choices among 3 goods

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1	300	$(10, 10, 10)$	$(10, 10, 10)$
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- Using local non satiation we can conclude the following.

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- Using local non satiation we can conclude the following.
- Since at prices $(10, 10, 10)$, the bundle $(15, 5, 9)$ costs less than $(10, 10, 10)$:

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- Using local non satiation we can conclude the following.
- Since at prices $(10, 10, 10)$, the bundle $(15, 5, 9)$ costs less than $(10, 10, 10)$:
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Example Continued

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1	300	$(10, 10, 10)$	$(10, 10, 10)$
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- Using local non satiation we can conclude the following.

- Since at prices $(10, 10, 10)$, the bundle $(15, 5, 9)$ costs less than $(10, 10, 10)$:
  - $(10, 10, 10) \succ (15, 5, 9)$.
- Since at prices $(10, 1, 2)$, the bundle $(9, 25, 7.5)$ costs as much as $(10, 10, 10)$:

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- Since at prices $(10, 1, 2)$, the bundle $(9, 25, 7.5)$ costs as much as $(10, 10, 10)$:
  - $(9, 25, 7.5) \succsim (10, 10, 10)$.
- Since at prices $(1, 1, 10)$, the bundle $(9, 25, 7.5)$ costs less than $(15, 5, 9)$.
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  - $(9, 25, 7.5) \succsim (10, 10, 10)$.
- Since at prices $(1, 1, 10)$, the bundle $(9, 25, 7.5)$ costs less than $(15, 5, 9)$.
  - $(15, 5, 9) \succ (9, 25, 7.5)$.
- Putting these together:

$$(10, 10, 10) \succ (15, 5, 9) \succ (9, 25, 7.5) \succsim (10, 10, 10)$$

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  - $(15, 5, 9) \succ (9, 25, 7.5)$.
- Putting these together:

$$(10, 10, 10) \succ (15, 5, 9) \succ (9, 25, 7.5) \succsim (10, 10, 10)$$

- These observations are not consistent with utility/preference maximization theory.

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Remark: The strict relation is derived from observation

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- $\succ^I$ is also defined directly from observation, and maybe different than the asymmetric component of $\succsim^I$.

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Exercise

Verify that $\mathbf{x}^j \succsim^R \mathbf{x}^k$ implies $\mathbf{x}^j \succsim^I \mathbf{x}^k$.
Verify that if $\mathbf{x}^j \succsim^R \mathbf{x}^k$ and $\mathbf{x}^k \succ^I \mathbf{x}^i$, then $\mathbf{x}^j \succ^I \mathbf{x}^i$.

Next Class

- Generalized Axiom or Revealed Preferences and Afriat's Theorem
- Production
- Firms' Choices